2.2.3 Polynomials and Polynomial Equations

In the case of ***equations***, which are not identities, we are interested in finding ***solutions.*** However, finding solution of a general *mathematical equation*, or of even a general *algebraic equation* is quite a complicated matter. Here, we restrict to *polynomial equations*, rather, to some very restricted types of polynomial equations.

Let us recall that a *polynomial* expression is a *mathematical* expression which may involve variables only with nonnegative integer powers. It may involve one or more variables. For example, $x^3 + 8x - 4$; $x^2 + (17)^{1/2}$ (the constant in a polynomial may have fractional power); $(x - 4y)^3$ etc. A polynomial equation is a mathematical expression involving exactly one relational operator, and that also must be equal-to sign (=) and two polynomial expressions, one on each side of '='. For example, $(x - 4y)^3 = x^2 + (17)^{1/2}$; $(x + y)^2 = 0$.

Some basic definitions around polynomial expression/equation

a) **Term:** each component of an expression separated by either of the two symbols '+' and '–'is called a term. For example, the expression $x^3 + 8x - 4$ has three terms, viz. x^3, 8x, and – 4 *(or, instead of – 4, we may take just 4).* However, $(x - y)^2$, at highest level, has only one term. Of course, on expansion, $(x - y)^2 = x^2 + y^2 - 2x\,y$ has 3 terms.

   Examples of power of a variable in a term*:* The expression$x^3y + 8x - 4y^2$ *has three terms, viz.* x^3y, *8x, and* $-4y^2$. ***The power of x in term*** x^3y *is 3; in term 8x is 1; and in* $-4y^2$*, it is 0. The* ***power of y in*** *term* x^3y *is 1; in 8x it is 0; and in term* $(-4y^2)$*, it is 2.*

   A ***single term***, at highest level, does not contain any of the two symbols '+' and '–'. A ***term of a polynomial expression/ equation*** involves only multiplication (×) including power/ exponentiation, e. g. $(45\, x^2\, y^3)$. However, a term of an ***algebraic expression/ equation*** may involve both (×) and division (÷) symbols, but none of '+' and '–'. For example, $(45\, x^2) / (78\, y^3)$ is an *algebraic* term, but not a *polynomial* term. ***Unless mentioned otherwise, our term is a polynomial term only.***

   Factor & Coefficient: The term, for example $(45\, x^2\, y^3)$ has many **factors**. For example. each of 3, 5, 9, 45, x, x^2, y, y^2, y^3, and any products of these is a factor of $(45\, x^2\, y^3)$. The *constant 45* is called ***coefficient*** in the term $(45\, x^2\, y^3)$. ***Similarly,****17* is the coefficient in the term (17xy) and the coefficient in the term –25x is –25.

b) **Monomial, binomial, polynomial:** Coming back to general ***polynomial*** expression**,** an expression**,** e. g. $(45\, x^2\, y^3)$**,** which involves only one term is called ***monomial*** expression; an expression**,** e. g. each of $(45\, x^2\, y^3 - 25)$ and ***a+ b,*** which involves two terms is called ***binomial*** expression; an expression**,** e. g. each of $(45\, x^2\, y^3 - 25x^4 + 71\, y^3)$ ***and a + b****– 72,* which involves three terms is called **trinomial** expressionand so on. In general, an expression containing, one or more terms—each with non-zero coefficient and variables having non-negative exponents—is called a **polynomial**. A polynomial may involve:

- one variable, e. g. $45\ x^3 - 25x^4 + 91\ x - 36$, involving only variable x,
- two variables, e. g. $(45\ x^2\ y^3 + y^2 - 25)$, involving two variables, viz. x and y,
- or more variables e. g. $x + 3y + z + 2u + v$ etc.

General form of a polynomial equation: For all practical purposes, a polynomial ***equation,*** containing at least one variable***,*** *can be* obtained by postfixing **'= *0*'** to a polynomial ***expression***, which does not already contain **'='*.*** For example, $(45\ x^2 y^3 + 71\ y^3 - 25x^4) = 0$*.*

Degree of a polynomial and a polynomial equation: For this purpose, ***degree of a term*** is the sum of the powers of all the variables occurring in the term, e. g. degree of the term $(-45\ x^2\ y^3)$ = power of x + power of y =2 +3=5; and, ***degree of the term (345) is 0***, as 345 = 345× 1 may be considered as $(345\ x^0)$, or $(345\ x^0\ y^0)$ etc. *(as $x^0 = 1$, for any x, which may be any variable or any non-zero constant)*. **Degree of a polynomial** is the *maximum* of the degrees of the terms occurring in the polynomial, e. g. degree $(45\ x^2\ y^3 + 71\ y^3 - 25x^4)$ is 5*.* Generally, a polynomial is written in order of decreasing degree of its terms: Though it is perfectly correct to write a polynomial as $45\ x^2\ y^3 + 71\ y^3 - 25x^4$,yet for better understanding, it is generally written as$45\ x^2\ y^3 - 25x^4 + 71\ y^3$.**Degree of a polynomial equation** is the *maximum* of the degrees of the terms occurring in the equation, e. g. ***degree of polynomial equation*** $(71\ y^3 + 45\ x^2\ z^2 y^3 + 2xy = -25x^4)$is *7.*

Name for a polynomial*:* Instead of repeating, if required, a polynomial expression at several places, we may give a name to it. If a polynomial, say $45\ x^2\ y^3 + 71\ y^3 - 25x^4 + 76$, involves two variables say x and y, then we may name it as ***P(x, y),*** with $P(x, y) = 45\ x^2\ y^3 + 71\ y^3 - 25x^4 + 76$.

Evaluating a polynomial for given values of involved variables: By definition of a variable, its value may change. The variable may take any value from a set of possible values, where each value of a variable is a constant of *appropriate* type. For example, a *real* variable may take any *real number* as its value, e. g. if x is a *real* variable, then each of 4.75 and √2 is a possible value of x, but 2+ 3 i, where $i^2 = -1$, is not a possible/appropriate value of x. For given (constant) values of the variables, e. g. x and y, the value of a polynomial, say P (x, y) is also a constant obtained by replacing the variables by respective values in the polynomial expression. For example, for *x=2 and y =* –1, ***the value of the polynomial*** $P(x, y) = 4\ x^2 - 7\ xy^2 + 11$is

$P(2, -1) = 4(2)^2 - 7(2)(-1)^2 + 11 = 16 - 14 + 11 = 13$, but

$P(-1, 2) = 4(-1)^2 - 7(-1)(2)^2 + 11 = 4 + 28 + 11 = 43$.

Solution of an (polynomial) equation: We have already mentioned that a polynomial equation—in say two variables x and y—is of the form

P (x, y) = Q (x, y) … … … (i)

where P and Q are polynomials in x and y. Then, (constant) values *a* of variable x and *b* of variable y constitute a solution of equation (i) just above, if values *P(a, b) and Q(a, b)* are equal.

2.2.4 Polynomial Identities

We have already discussed that $x^2 - 4 = 0$ is a polynomial equation, having ***two*** solutions, viz. x = 2, and x= –2. However, the equation $x^2 - a^2 = (x+ a) (x- a)$ is satisfied by ***every value of*** x.

Equations like $x^2 - a^2 = (x+ a) (x- a)$, *which are satisfied by every value of variable x*, are called **identities,** and, whenever the distinction between equation and identity needs to be highlighted, then an identity is distinctly written, by using ≡**,** as

$$x^2 - a^2 \equiv (x+ a) (x- a).$$

There are a number of well-known polynomial identities, including the following ones. Though these are *identities*, we are using the symbol *'='* instead of '≡'.

1) $(x + y)^2 = x^2 + y^2 + 2xy$
2) $(x - y)^2 = x^2 + y^2 - 2xy$
3) $(x^2 - y^2) = (x + y) (x - y)$
4) $(x + a) (x + b) = x^2 + (a + b) x + ab$
5) $(x + y + z)^2 = x^2 + y^2 + z^2 + 2xy + 2xz + 2yz$

By replacing throughout any one or two of x, y and z by corresponding negative(s), e. g. by replacing y by (– *y) etc.*, we get some other identities. For example, by replacing y by (– y), and z by (–z), we get

6) $(x - y - z)^2 = x^2 + (-y)^2 + (-z)^2 + 2x (-y) + 2x (-z) + 2(-y) (-z)$

   i. $= x^2 + y^2 + z^2 - 2x y - 2xz + 2yz$

7) $(x + y)^3 = x^3 + y^3 + 3xy (x + y)$

Also, by replacing y by (–y) in the above Identity, we get

8) $(x - y)^3 = x^3 + (-y)^3 + 3 x (-y) (x + (-y)) = x^3 - y^3 - 3 xy (x - y)$
9) $x^3 + y^3 + z^3 - 3xyz = (x + y + z) (x^2 + y^2 + z^2 - xy - yz - zx)$

Applications of Polynomial Identities

Example: Evaluate 105 × 95 without multiplying directly.

Ans**.** $105 \times 95 = (100 + 5) (100-5) = 100^2 - 5^2$ *(using $(x^2 - y^2) = (x + y) (x - y)$)*

= 10000– 25= 9975.

Example: Factorize $(625/16)\,x^2 - (9/49)\,y^2$

Ans: As $(625/16) = (25/4)^2$ and $(9/49) = (3/7)^2$, therefore,

$(625/16)\,x^2 - (9/49)\,y^2 = (25/4)^2\,x^2 - (3/7)^2\,y^2$

$= [(25/4)\,x]^2 - [(3/7)\,y]^2$

$= [(25/4)\,x + (3/7)\,y]\,[(25/4)\,x - (3/7)\,y]$

Check Your Progress 2

1) Write $(2a - 4b - 5c)^2$ in expanded form

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2) Factorize $(625/16)\,x^4 - (9/49)\,y^4$, with factors having only real coefficients.

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3) Factorise $x^2 + 4y^2 + z^2 - 4xy - 4yz + 2xz$.

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2.2.5 Algebra of Polynomials

In this subsection, we discuss how to add, subtract, multiply and divide polynomials. First, we take up addition and subtraction of polynomials.

1) **Sum and Difference of Two Polynomials**

For this purpose, for a short while, we shift our attention from the *abstract world of numbers, constants and variables etc*. to the *concrete world of animals*. Sum of 3 cows and 4 cows may be written in simplified form as (3 +4) cows, i. e. as 7 cows. But *sum of 3 cows and 4 horses* can NOT be simplified in the similar way; the sum of 3 cows and 4 horses can, at most, be written in simplified form as *3 cows + 4 horses.* Thus, if *c* represents a *cow* and *h* a *horse*, then sum of 3c and 4c can be simplified as 7c. But simplified form of the sum of 3c and 4h is only *3c+ 4h*.

Returning to the abstract world of numbers now. ***Each of x, x^2, x^3, y, y^2, xy, xy^3, and a constant c or 3 is a distinct animal***. Hence, the sum $3x^2 + 4x^2$ can be simplified as $7x^2$. But the sum $3x + 4x^2$ cannot be simplified in similar way. Of course, $3x + 4x^2$ may be further simplified as $x(3 + 4x)$.

Like terms & Unlike terms: For finding sum of two mathematical expressions—similar to the categorization of animals—we categorize terms occurring in the expressions. For our purpose, we categorize terms as ***'Like terms'*** and ***'Unlike terms'. 'Like terms'*** *are sort of 'animals of the same type' and hence their sum can be simplified. Two* ***unlike terms*** *are sort of 'animals of different types', and hence, their sum cannot be simplified in the same manner.*

Like Terms are terms which are identical except for their constant parts. For example, ***two terms are 17 $x^2 y^3$ and –39 $x^2 y^3$ are like terms***, their sum can be written in the simplified form as $(17 + (-39))\ x^2\ y^3 = -22\ x^2\ y^3$. However, the ***two terms 17 $x^2 y^2$ and –39 $x^2 y^3$ are unlike terms***, the sum of $17\ x^2\ y^2$ and $-39x^2\ y^3$ cannot be simplified in similar manner. Their sum has to be written as ***$17\ x^2\ y^2+(-39\ x^2\ y^3)$, or as $17\ x^2\ y^2-39\ x^2\ y^3$.*** Of course, the sum***$17\ x^2\ y^2-39\ x^2\ y^3$may be simplified*** further in a different manner as $x^2\ y^2\ (17-39y)$. As addition and subtraction are opposite operations, therefore the above argument equally applies to subtraction also.

Example of sum of two polynomials:

$P(x, y) = 3x^2- 5xy +3x +8$

$Q(x, y) = 6y^2 + 12\ xy - 5x +16$ i. e.

$P(x, y) + Q(x, y) = (3x^2- 5xy +3x +8) + (6y^2 + 12\ xy - 5x +16)$

It may be simplified as

$(3x^2- 5xy+ 12\ xy +3x - 5x +8+16) + 6y^2$

$= 3x^2 + (- 5+ 12)\ xy +(3 - 5)x +(8+16) + 6y^2$

$= 3x^2 + 7\ xy + 6y^2 - 2x + 24.$

Example of difference of two polynomials:

In order to find the difference **P (x, y) – Q (x, y), first,** we get a new polynomial R(x, y) from Q(x, y), by changing each '+' before a term to '–', and each '–' before a term to '+', and keeping the rest of the polynomial expression unchanged. Then we find **P (x, y) + R (x, y).**

For example, to find P(x, y) – Q(x, y) for $P(x, y) = 3x^2- 5xy +3x +8$ *and* $Q(x, y) = 6y^2 + 12\ xy - 5x +16$, first we obtain $R(x, y) = - 6y^2- 12\ xy + 5x - 16$. And then calculate P(x, y) + R (x, y) as above.

2) **Product of Two Polynomials**

Sum and difference do NOT produce new kinds of terms *(each term is a kind of animals)*. But product and quotients generally produce new kinds of terms.

Simplest case is that in which one of the polynomials is a constant. For example,

$P\ (x, y) = 3x^2- 5xy +3x +8$; and $Q\ (x, y) = 7$, then $P \times Q = (3x^2 - 5xy + 3x + 8) \times 7$

$= 3\times 7x^2 - 5\times 7xy + 3\times 7x + 8\times 7$

$= 21\,x^2 - 35\,xy + 21\,x + 56.$

Product of two general polynomials is obtained from *products of the terms*, one from each of the two polynomials as follows: If polynomials $P = T_1 + T_2 + \ldots + T_m$ and $Q = S_1 + S_2 + \ldots + S_n$ be two polynomials, respectively having terms $T_1, T_2, \ldots T_m$ and $S_1, S_2, \ldots, S_n$. Then the polynomial product, $P \times Q$ is ***the sum of term-products*** $T_i \times Sj$ where $i = 1, 2, \ldots, m$ and $j = 1, 2, \ldots, n$.

Term-products $T_i \times Sj$ is explained through the following example:

If $T = 5\,x^2\,y^3$ and $S = -16\,x\,y^2\,z^2$, then

$T\times S = (5)\,(-16)\,x^{2+1}y^{3+2}z^{0+2}$

$= -80\,x^3\,y^5\,z^2$,

i.e., constants are multiplied as usual, and powers of each of the variables occurring in the two terms are added to obtain product-term.

Example of product of two polynomials:

Find product of $P(x, y) = (4x - 3y + 5)$ and $Q(x, y) = (2x - 7)$

One of the methods may be first to take product of each term of $P(x, y)$ with whole of $Q(x, y)$

$= (4x - 3y + 5)\times Q(x, y)$

$= 4x \times Q(x, y) - 3y \times Q(x, y) + 5\times Q(x, y)$

$= 4x \times (2x - 7) - 3y\times(2x - 7) + 5\times (2x - 7).$

$= [4x \times 2x + 4x \times (-7)] + [-3y\times 2x - 3y\times(-7)] + [5\times 2x + 5\times(-7)]$

$= [8x^2 - 28x] + [-6xy + 21\,y] + [10x - 35]$,

Collecting ***like terms***, and simplifying, we get

$= 8x^2 - 28x + 10x - 6\,xy + 21\,y - 35 = 8x^2 - 18x - 6\,xy + 21\,y - 35.$

Check Your Progress 3

1) For $P(x, y) = 4x^2 - 3xy + 5y$, and $Q(x, y) = -4x^2 - 3xy + 5y^2$ find the (i) sum $P(x, y) + Q(x, y)$ and the (ii) difference $P(x, y) - Q(x, y)$.

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2) Find product $(4x^2 - 3xy + 5y) \times (2x^2 - 7y)$

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2.3 SOLVING POLYNOMIAL EQUATIONS

As mentioned earlier, finding solution of a general *mathematical equation*, or of even of a general *algebraic equation* is quite a complex matter. Even solving general *polynomial equations* is quite complex issue, and hence, we restrict to only the following *special types of polynomial equations*:

i) Single linear equations in one variable
ii) Single linear equation in two variables
iii) System of two or more linear equations in two or more variables
iv) Quadratic equation in one variable

Before we define the various concepts—like linear and quadratic equations etc.—involved in (i)-(iv) above, and go into details of solution processes, it may be pointed out that—in view of our earlier experience of solving such equations—treatment here is not at all exhaustive. The methods are illustrated only through some examples.

An equation is said to be ***linear*** in one, two or more variables, if each term involving variable of the equation has degree 1. For example, $3 – 29\ x = 5x +17$ (linear in one variable x); $7y – 29\ x = 5x +17$ (linear in two variables x and y); $x+2z – 7\ y =17$ (linear in three variables x, y, z)

None of the following is a linear equation: $3 – 29\ x^2 = 5x +17$ *(power 2 of x involved)*; $45= 27 + 18$ *(no variable is involved)*; $xy = x+ 55$; *(no products of variables can occur in a linear equation);* $\log x +17 = 0$; $e^x= x+ 25$; *(logarithmic and exponential functions cannot be involved in polynomial, and hence in linear equations).*

System of linear equations: Two or more linear equations—in two or more variables, say x and y—constitute a system of linear equations. We need to find values x =a, and y= b. Where λ for constants a and b are such that *each of the equations* involved in the system is satisfied *by the same values* x =a, and y= b. For example, each of the two equations $2x +3y = x + 11$; and $7x – 2y = 8$is satisfied by x =2 and y = 3.

An equation is said to be ***quadratic*** in ***one*** variable, say y, if one, or more terms in it are non-zero constant multiples of y^2, and other terms involve either constants or a constant multiple of y. *General form of a quadratic equation in y:* $a\ y^2+b\ y +c = 0$, *with* $a \neq 0$. For example, each of the following is a quadratic equation in one variable: $33\ y^2 + 3= 0$; $25x^2 = 5x^2 +17x – 52$.

None of the following is a quadratic equation in one variable:– $9y= 5 +17$; $45y= 27 + 18$

2.3.1 Solving Linear Equation in One Variable

A *general* linear equation in one variable, say x, is of the form

$a\ x + b = cx +d$ … … … (i)

where, a, b, c and d are constants, with $a \neq c$. The reason for requiring $a \neq c$ is that, in case of $a = c$, the equation (i) reduces to $b=d$, which does **not contain any variable.**

Basic rules in solving such equations are

i) to *transfer all constant terms* on one side of '=', generally to R. H. S., and all *terms containing variable* to the other side of '=', generally L. H.S.,

ii) while switching sides of a term, its sign is switched from + to –, and from – *to* +.

Example: Solve the *linear* equation $2x + 5 = 4 – 3x$

Ans: By the application of rules (i) & (ii) above to our equation, we get

$2x + 3x = 4 – 5$, implying

$5x = – 1$,

$x = – 1/5$, is the required solution.

Some equations are not linear, not even polynomial equations, but may be converted/ reduced to a linear equation, where the new equation is not exactly equivalent, but nearly equivalent to the original equation.

Example: Consider the equation

$(3x+2)/(2x-1) = 2.$ … … … (i)

The equation (i) above ***is not exactly equivalent to***

$(3x+2) = 2.(2x-1).$ … … … (ii)

because equation (i) is NOT defined at $x = ½$, but equation (ii) is defined at $x = ½$.

But except for the value x =1/2, equations (i) and (ii) are equivalent.

Now, equation (ii) is linear,

For solving equation (ii), we shift all x's to L.H. S. and the rest of the terms to R.H.S., we get

$3x – 4x = –2 –2 = –4$, implying

$– x = –4$, or $x= 4$.

Thus, $x= 4$ is a solution of (ii), and hence, is a solution of equation (i) also, as $4 \neq ½$.

2.3.2 Single Linear Equation in Two Variables

A *general* linear equation in two variables, say x and y, is of the form

$Ax + By + C = Dx + Ey + F$ … … ……….. (i)

where, A, B, C, D, E and F are constants.

By appropriate shifting of terms, (i) may be rewritten as

(A–D) x + (B– *E)* y + (C – F) = 0

By taking a = (A–D), b = (B– *E), and c*= (F – C),

Thus, a general equation in two variables may be considered of the form

ax + by = c …………………(ii)

We consider some special cases:

Case (a) if a = 0 and b= 0 then the equation (ii) reduces to c=0, which is not linear, and even may not be true if c ≠ 0.

Case (b) if a = 0, but b≠ 0, then equation (ii) reduces to b. y= c, which is a linear equation in one variable, covered under case(i) above

Case (c) of if a ≠0, but b= 0 is similar to case(b) above.

Thus, in the following, we assume that a ≠ 0 and b ≠ 0. Then the equation (ii) reduces to

by = – a x + c

or y = [– a x + c]/ b…………………………. (iii)

For each value of x, we get a value of y. Therefore, the equation has **infinitely many** pairs of solutions.

Example: Solve the linear equation

3x – 2 y + 4 = 4x + 3 y – 5 ……… ………………… (iv)

In such cases, we transfer all terms containing y on one side of '=', and all other terms on the other side, taking care that while switching sides, signs of terms get changed from '+' to '–', and from '–' to '+'. Thus, we get

– 2y – 3y = 4x – 3x – 5 – 4, i.e.

– 5y = x – 9

or, y = (9 – x)/ 5,\

We get infinitely many pairs of values of (x, y) satisfying the equation (iv), including the following (1, 8/5), (2, 7/5), (3, 6/5, (4, 1), …

2.3.3 System of Two or More Linear Equations in Two or More Variables

In its general form, the topic will be discussed in detail in later units, specially, in units dealing with matrices and determinants. Here we discuss only a ***special case*** when there are TWO linear equations in TWO variables.

A general linear equation in two variables may be considered of the form

ax + by = c

Hence, a general system of two linear equations is of the form

$ax + by = c$ … … …………(i)

and

$dx + ey = f$ … … …………(ii)

One of the approaches to solving this system of equations is to get a new equation not involving one of the variables, say x, by multiplying equation (i) throughout by d, and equation (ii) throughout by (– a), and then getting the new equation by adding corresponding terms.

The rest of the method is explained through attempting to solve the following system of equations

$3x - 5y = -9$ … …………………. (iii)

$2x + 3y = 13$ … …………………. (iv)

Multiplying coefficients of terms in (iii) by 2 throughout (where 2 is coefficient of x in (iv)), we get

$6x - 10y = -18$ … …………………. (iii')

Similarly, multiplying coefficients of terms in (iv) by 3 throughout (where 3 is coefficient of x in (iii), we get

$6x + 9y = 39$ … …………………… (iv')

By subtracting coefficients of terms in (iv') from the corresponding ones of (iii'), we get

$-19y = -18 - 39 = -57$

$y=3$

Substituting y=3 in (iii) (we may do so even in (iv), (iii'), and (iv')), we get

$3x - 5(3) = -9$

$3x = 15 - 9 = 6$

$x = 2$.

Thus x=2, and y=3 is the **unique solution** of the system of equations given by (iii) & (iv).

2.3.4 Quadratic Equations

From our earlier discussion, we know that, for variable x, equation $ax^2+bx+c=0$, ***with*** $a \neq 0$ is the general form of a quadratic equation.

A quadratic equation has two solutions, not necessarily distinct, given by

i) $x = [-b + \sqrt{(b^2 - 4ac)}] / (2a)$, and

ii) $x = [-b - \sqrt{(b^2 - 4ac)}] / (2a)$

If $(b^2 - 4ac) = 0$, then the two roots coincide.